

Integrating the Mashreq Gas Market and the European Internal Gas Market

“Gas exports from Egypt by Pipeline & LNG”

*By
Ahmed Said El Ashmawi
Head of the Minister's Technical office
Ministry of Petroleum-Egypt*

19th of June 2008

Energy as a driving force for Integration

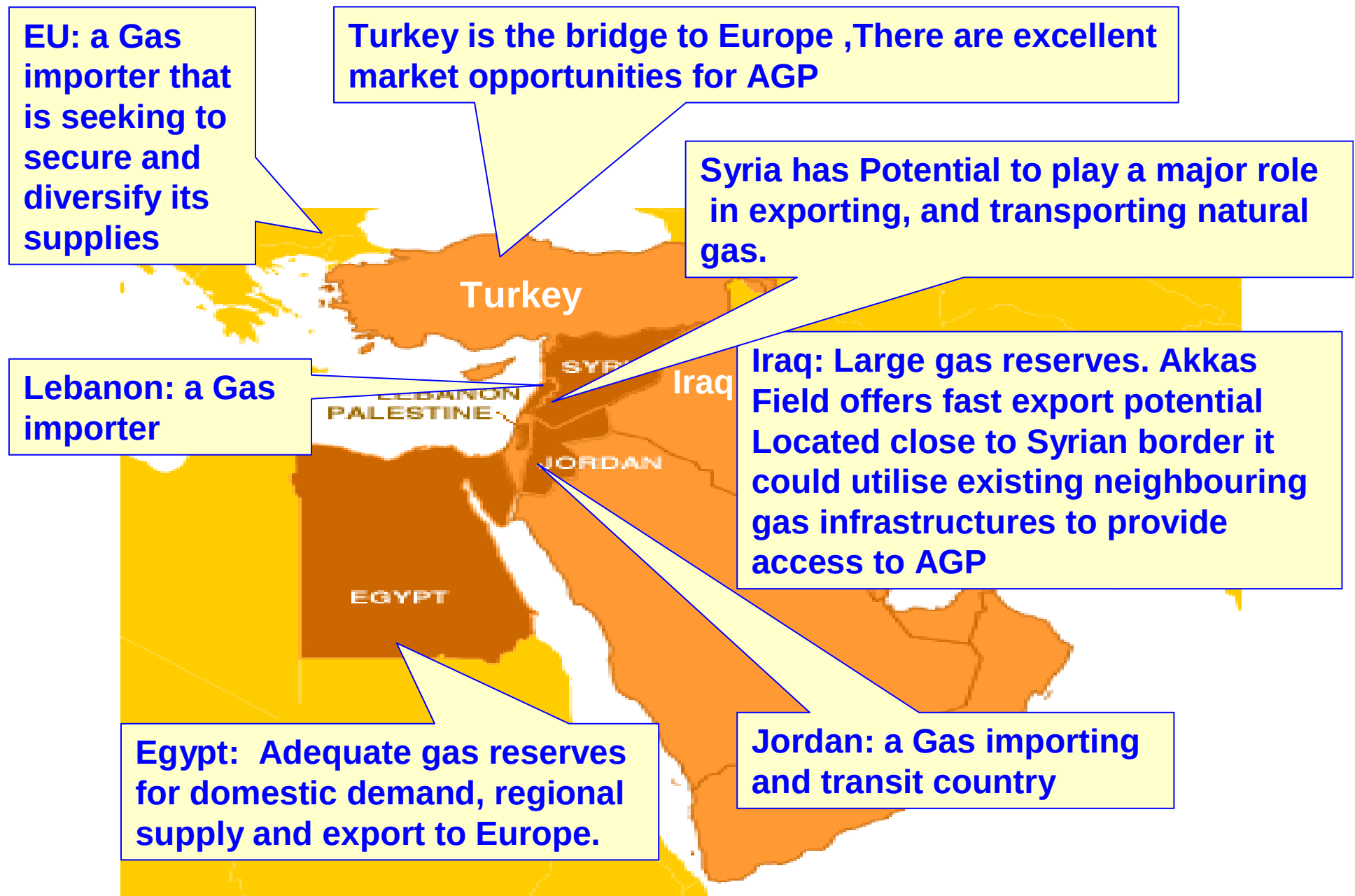
- The European “Coal & Steel Treaty” has launched the European community integration in 1952, and was very largely a “Peace project” between France and Germany and provided a model for promoting social change through a new form of co-operation at European level.

Energy was thus at the heart of the original idea of European integration and still covers a crucial role in the European policy today.

Integration of the Euro-Mashreq Gas Market

In the same manner, The establishment of an interconnected Mashreq gas market, that could be further extended to Europe, can play an important role as a vehicle to enhance the security and safety of energy supplies, and in the same time promote regional cooperation and integration.

Features of the Euro Mashreq Gas Market



Features of the Mashreq Gas Market

What Does It Take To Establish A Regional Gas Market

1. A physical infra structure to link buyers and sellers
2. A legal and Regulatory Frame work that allows the Market to operate smoothly and attract investors
3. Qualified technical expertise to operate safely, efficiently, and according to industry standards

Euro-Mashreq gas Cooperation Center

“To contribute to the integration of the gas markets of Egypt, Jordan, Lebanon & Syria in view of creating a regional internal gas market to be integrated with the EU Internal Gas Market”

Timing

33 months, from 1st February 2006 to 31st October 2008

Budget

€7 million, of which EU grant €6 million, beneficiary countries €1 million in kind contributions

Location

Egypt, Jordan, Lebanon, Syria

Mechanism

The Euro-Arab Mashreq Gas Co-operation Centre, headquartered in Damascus

Approach

Gas Master Plan, Legislation/Regulation, Know-How Transfer



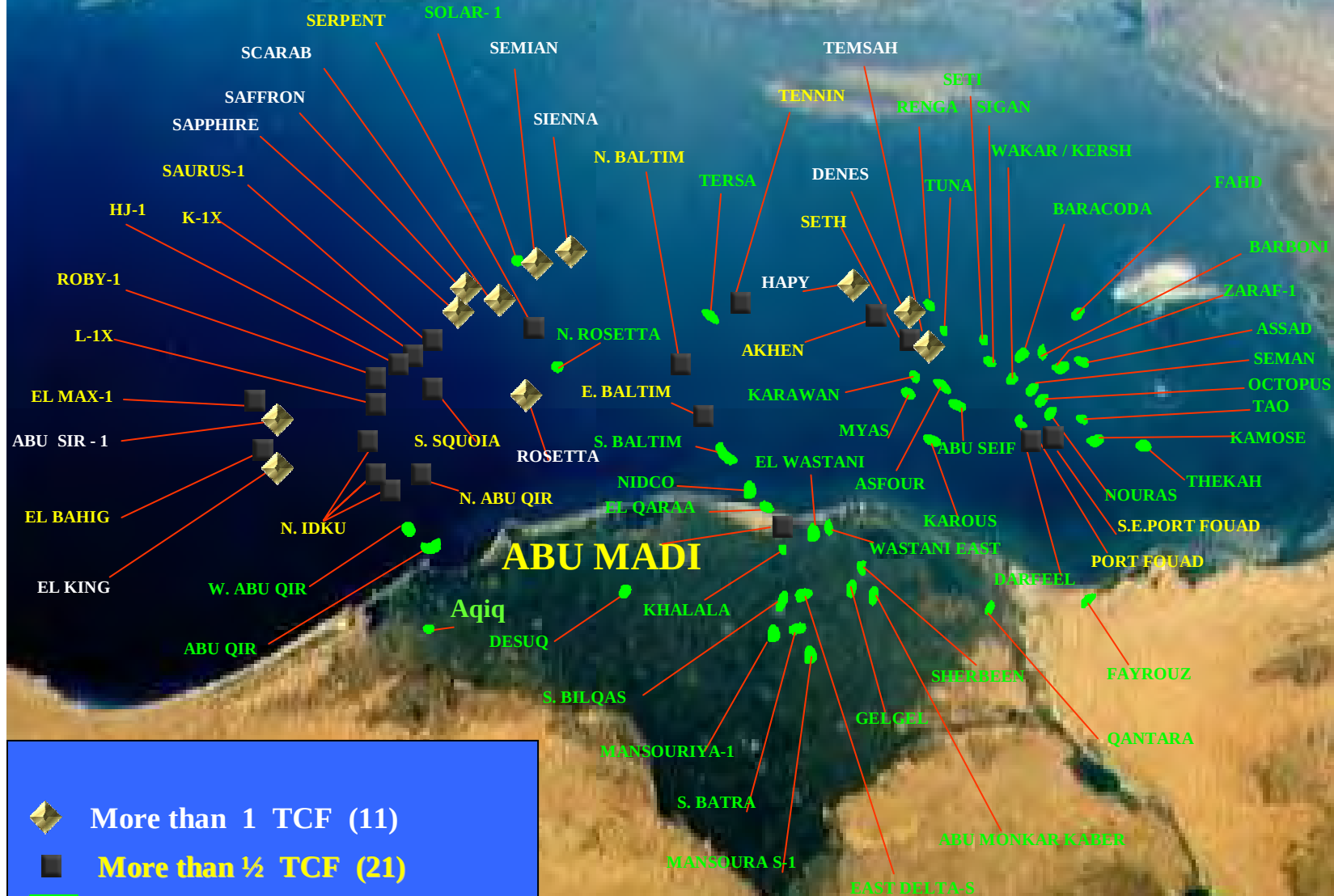
Overview of Egypt's Oil & Gas Industry



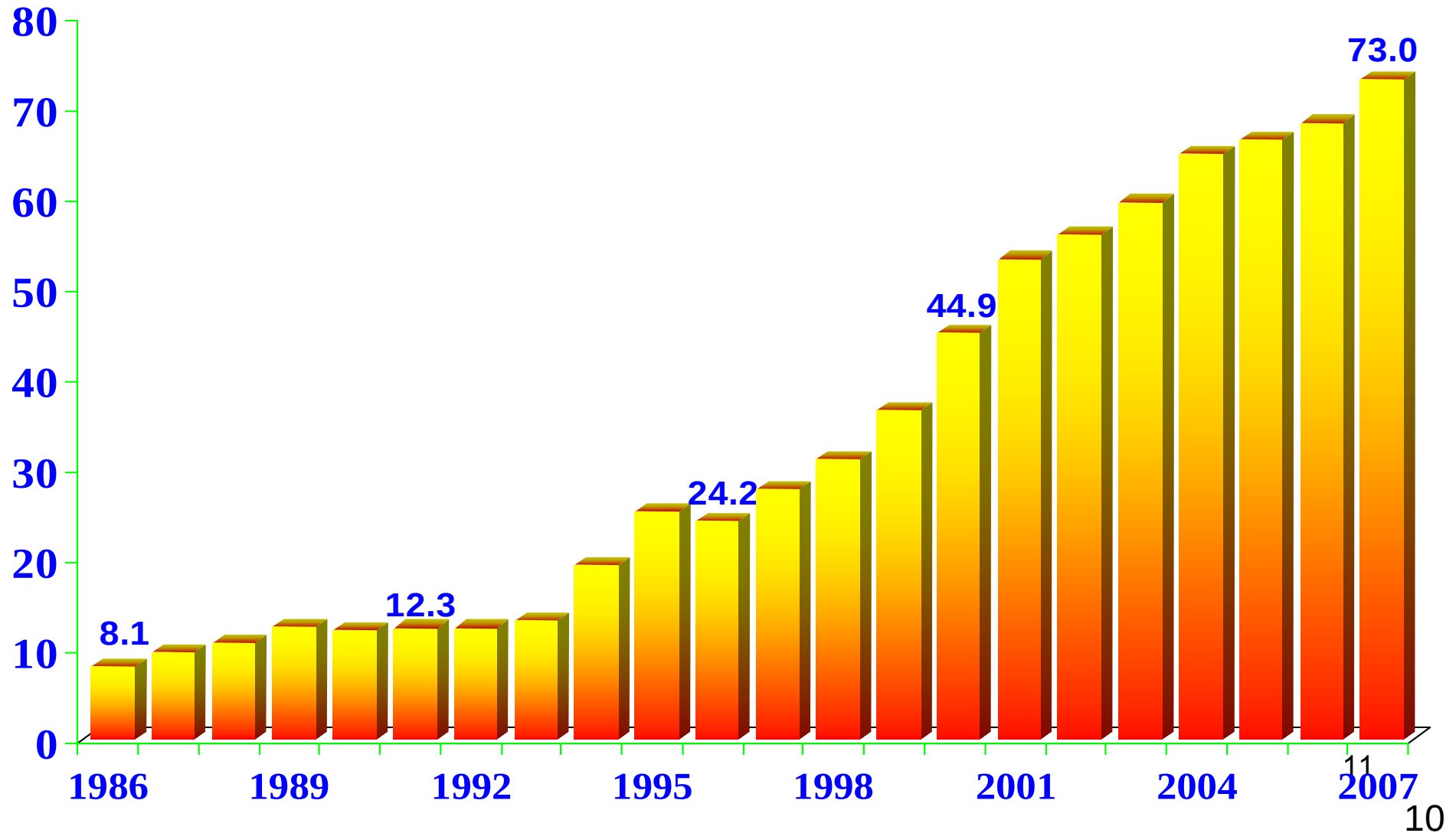
Overview of Egypt's Oil & Gas Industry

- Egypt is one of the world pioneers to start oil activities.
- First discovery of crude oil in Egypt in recent history was in 1868. The first oil well drilled in Egypt in 1886, and the first commercial production started in 1909.
- Morgan Field, the first major oil field in Egypt, was discovered by Amoco in 1963.
- The first gas field “Abu-Madi” was developed in 1975.

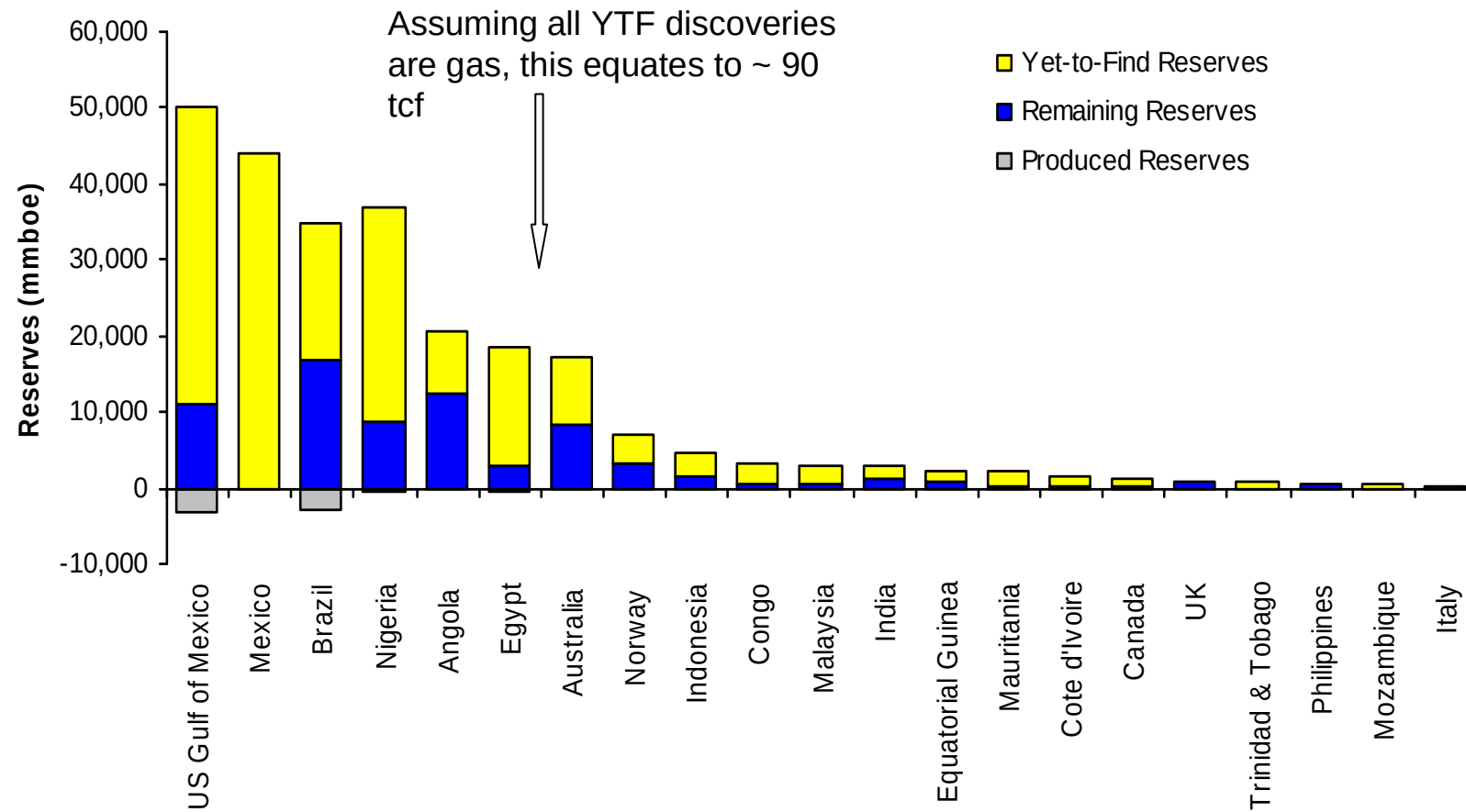
Egypt Offshore Mediterranean ... A World Class Gas Basin



Egypt Proven Gas Reserves (TCF)



Yet to Find (YTF) – Deepwater Potential Only





Diversified Natural Gas Market



Egypt among the First Twenty Countries in the World In Natural Gas Production

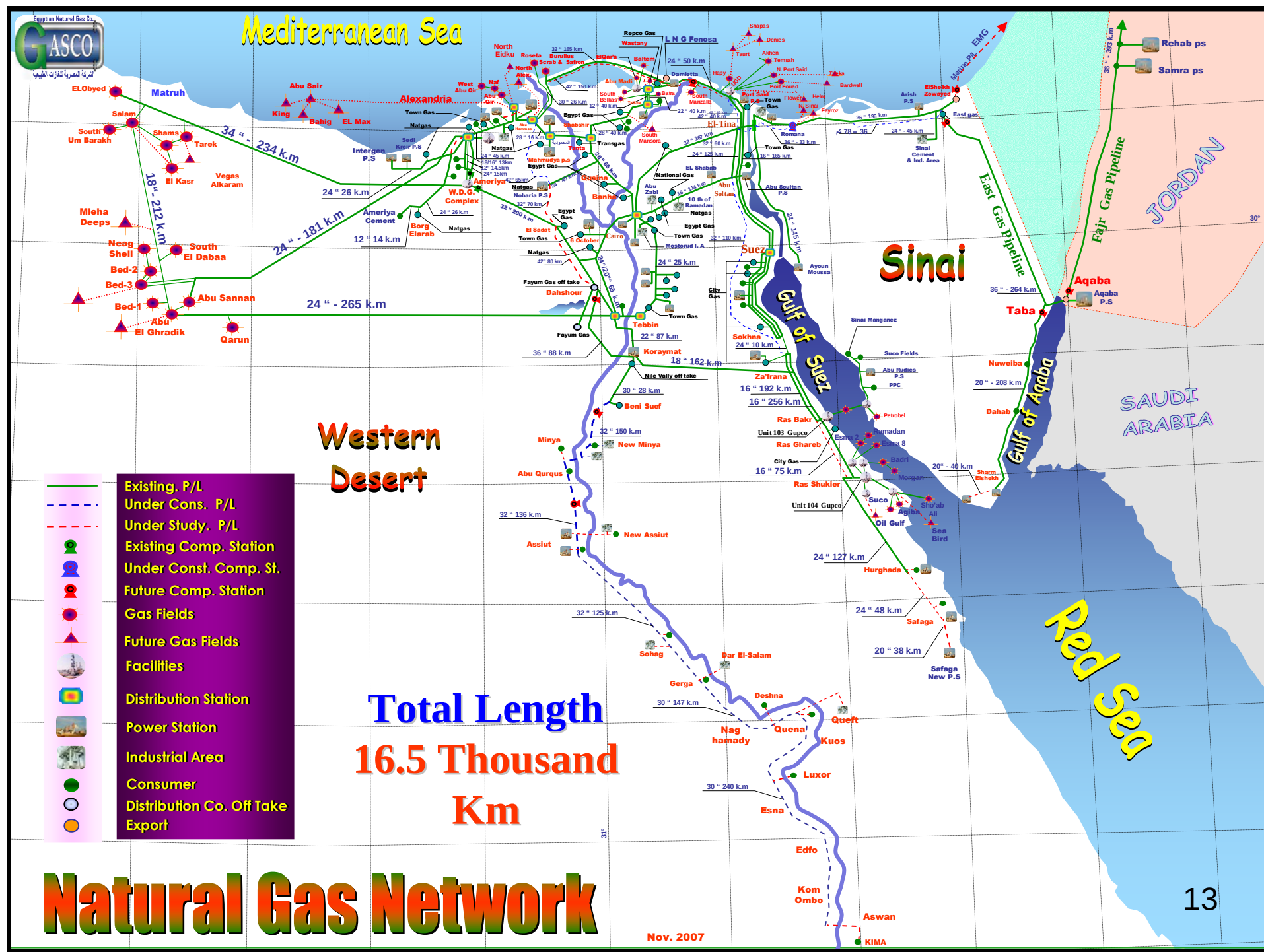
World Oil and Gas Review 2007

Natural Gas

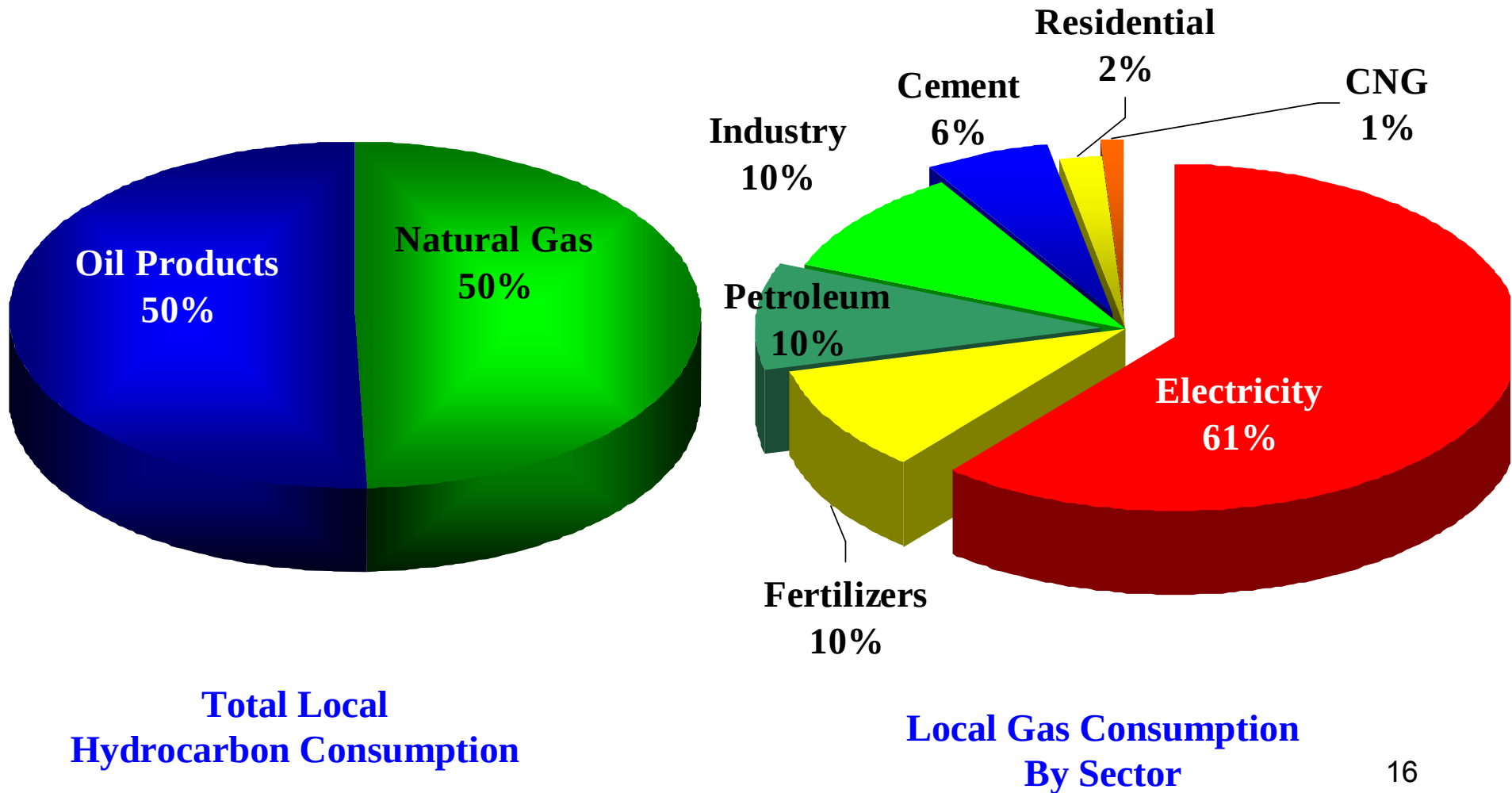
Production ■ Produzione

The first twenty countries in the world □ Primi venti paesi del mondo
(billion cubic metres - miliardi di metri cubi)

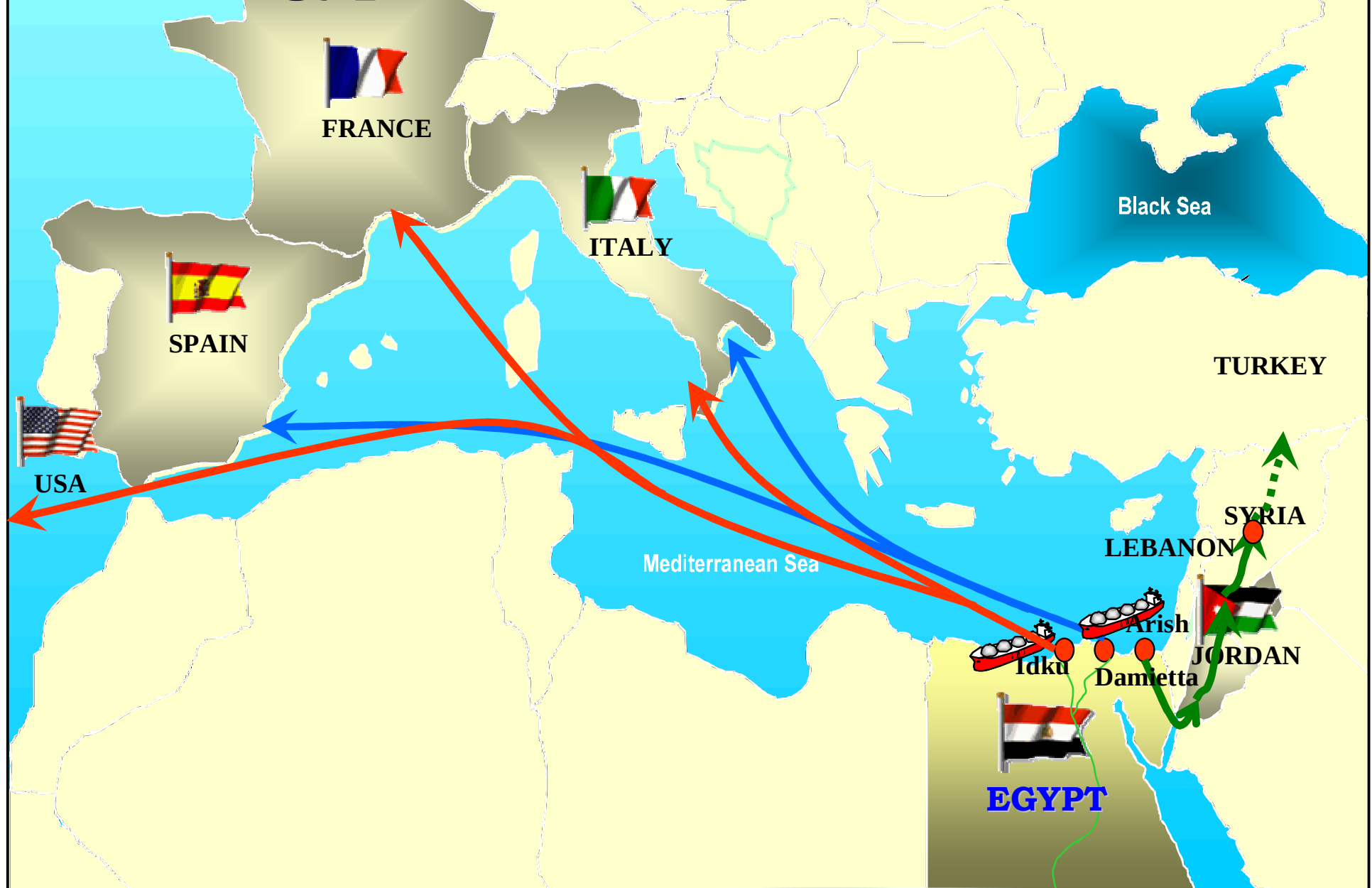
	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005
First twenty	1,909.91	1,991.43	1,980.56	2,029.61	2,092.05	2,158.56	2,189.03	2,230.52	2,308.66	2,358.19	2,401.39
Russia	573.13	579.06	549.67	569.08	569.18	561.43	559.07	571.93	596.13	607.44	614.25
United States	519.75	525.11	527.47	529.14	525.90	534.35	547.88	527.71	534.52	522.98	508.28
Canada	155.35	161.28	164.20	169.32	172.29	176.94	181.64	182.88	180.29	179.75	182.80
Algeria	61.83	65.59	75.61	80.84	90.80	92.58	87.10	89.64	94.69	93.35	101.57
United Kingdom	75.99	90.51	92.19	96.81	106.38	116.35	113.61	111.40	110.48	103.05	93.94
Norway	33.70	43.85	48.92	49.32	52.64	54.64	58.31	70.57	79.04	83.99	90.29
Iran	42.59	42.31	46.77	49.39	56.34	58.98	61.53	69.76	78.22	81.99	84.02
Indonesia	69.97	73.88	76.53	74.57	80.96	75.17	72.94	78.58	82.19	79.83	83.32
Saudi Arabia	41.82	43.37	44.66	45.52	44.99	48.39	52.30	55.84	58.51	63.99	67.71
Netherlands	72.10	81.51	72.27	68.71	64.07	61.90	66.45	64.73	62.27	73.45	67.32
China	22.14	24.82	25.57	28.72	31.09	33.56	37.42	40.30	43.20	52.04	61.90
Turkmenistan	34.68	34.00	16.74	12.81	22.18	45.58	49.83	51.97	57.47	57.31	61.61
Uzbekistan	45.49	45.72	47.39	53.22	53.99	54.79	55.77	56.18	56.34	57.31	58.75
Malaysia	26.41	32.93	37.50	37.90	39.30	45.65	44.43	45.82	47.96	55.43	58.60
Argentina	25.44	27.79	30.60	32.32	36.16	40.19	39.95	38.37	42.66	46.67	45.77
Qatar	8.76	9.11	12.60	15.06	22.84	27.35	28.19	30.96	32.94	41.17	45.69
United Arab Emirates	29.92	34.63	34.69	35.41	36.34	38.06	37.60	41.45	42.79	44.31	44.56
Australia	29.82	30.60	30.48	31.68	32.80	34.04	34.73	36.19	37.34	38.17	44.34
Mexico	28.76	32.68	33.70	36.39	36.85	37.39	37.68	38.64	40.50	42.34	44.06
Egypt	12.27	12.69	13.01	13.39	16.96	21.22	22.60	27.61	31.12	33.62	42.61
Rest of the World	275.77	287.93	298.76	305.01	312.70	334.26	344.79	358.21	383.51	410.03	436.07
World	2,185.68	2,279.36	2,279.32	2,334.63	2,404.74	2,492.82	2,533.81	2,588.73	2,692.17	2,768.22	2,837.46



Gas Share in the Local Market 2006/2007



Egypt's Gas Export Projects



Damietta LNG Complex

- Investments
- Markets
- Shareholders

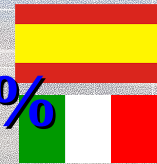
- Capacity
- Start-Up
- General Contractor
- Subcontractor

1300 MMUS\$ (including LNG Tankers)

Spain/Italy

Union Fenosa Gas

80%



EGAS/EGPC

20%



7.5 BCM/Year

Jan. 2005

Kellogg / Brown&Root



Enppi/Petrojet

Idku LNG Complex

- Investments
- Markets
- Shareholders

- Capacity
- Start-Up

- General Contractor
- Sub Contractor

1900 MMUS\$ (2 Trains)

France/ USA/Italy

BG 35.5%



Petronas 35.5



EGAS/EGPC 24%



Gas De France 5%



10.0 BCM/Year

May-2005 (1st Train)

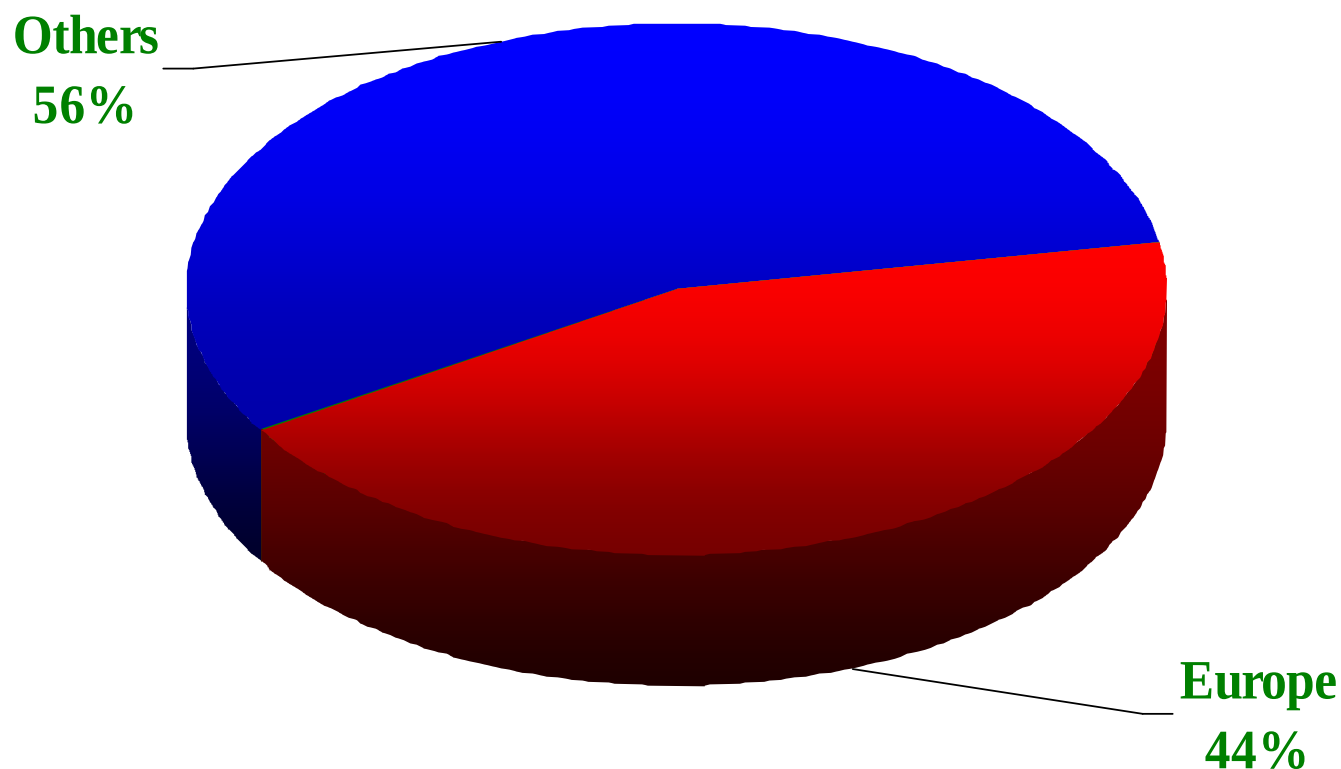
Sep-2005 (2nd Train)

Bechtel



Enppi/Petrojet

Shipments of Egyptian LNG Exported to Europe January 2005 – End of May, 2008



Total Shipments of LNG Exported = 450 shipments

The Arab Gas pipeline

The Arab Gas Pipeline

The Arab Gas Pipeline, aims to transport and distribute Egyptian Natural gas to Jordan, Syria and Lebanon in its first phases. The project can be further expanded to allow Iraq and other Major gas producers in the region to export their natural gas to Europe, or even in the opposite direction through additional networks and loops thus forming the “Arab Gas Network”.

Main Milestones for the Arab Gas Pipeline



(December 2000) : Signature of the MOU between Egypt, Syria and Lebanon to construct the Arab Gas Pipeline

(February 2001) : Jordan joins the Agreement

(June 2001) : Signature of Gas Sales Agreement between Egypt and Jordan

(September 2001) : Arab Funding agents agree to finance the Project



Arab Gas Pipeline Phases:

- Over all length= 1500 km
- Nominal capacity: 10 BCM/Y

- ➔ **Phase-1 Arish-Aqaba**
(264 km) *July 2003*
- ➔ **Phase-2 Aqaba-Rehab**
(394 km) *February 2006*
- ➔ **Phase-3 Rehab-Syrian Borders** (30 km) *completed early 2008*
- ➔ **Phase-4 Rehab-Homs** (330 km) *completed early 2008)*
- ➔ **Phase-5 Homs-kilis** (220 km) *(under negotiation)*
- ➔ **Phase-6 Kilis-Turkish Network** (90 km) *(under tendering)*
- ➔ **Lebanon Branch** *completed*

SURİYE-TÜRKİYE DOĞAL GAZ BORU HATTI

Responsibility :Botas

- 36" dia Pipeline (Kilis-Türkoğlu Interconnector)
- Length 94 km

Status: Under Realization

Arab Gas Pipeline



BOTAŞ

BORU HATLARI İLE PETROL TAŞIMA A.Ş.
ANKARA - TÜRKİYE

- SURİYE-TÜRKİYE DOĞAL GAZ BORU HATTI
- BTC HATTI
- SİVAS-MERSİN DOĞAL GAZ BORU HATTI
- NABUCCO PROJESİ İRAK DOĞAL GAZ BORU HATTI
- İRAK-TÜRKİYE VE BATMAN DÖRTYOL HAM PI BORU HATLARI

Gas to Europe: Nabucco and ITGI

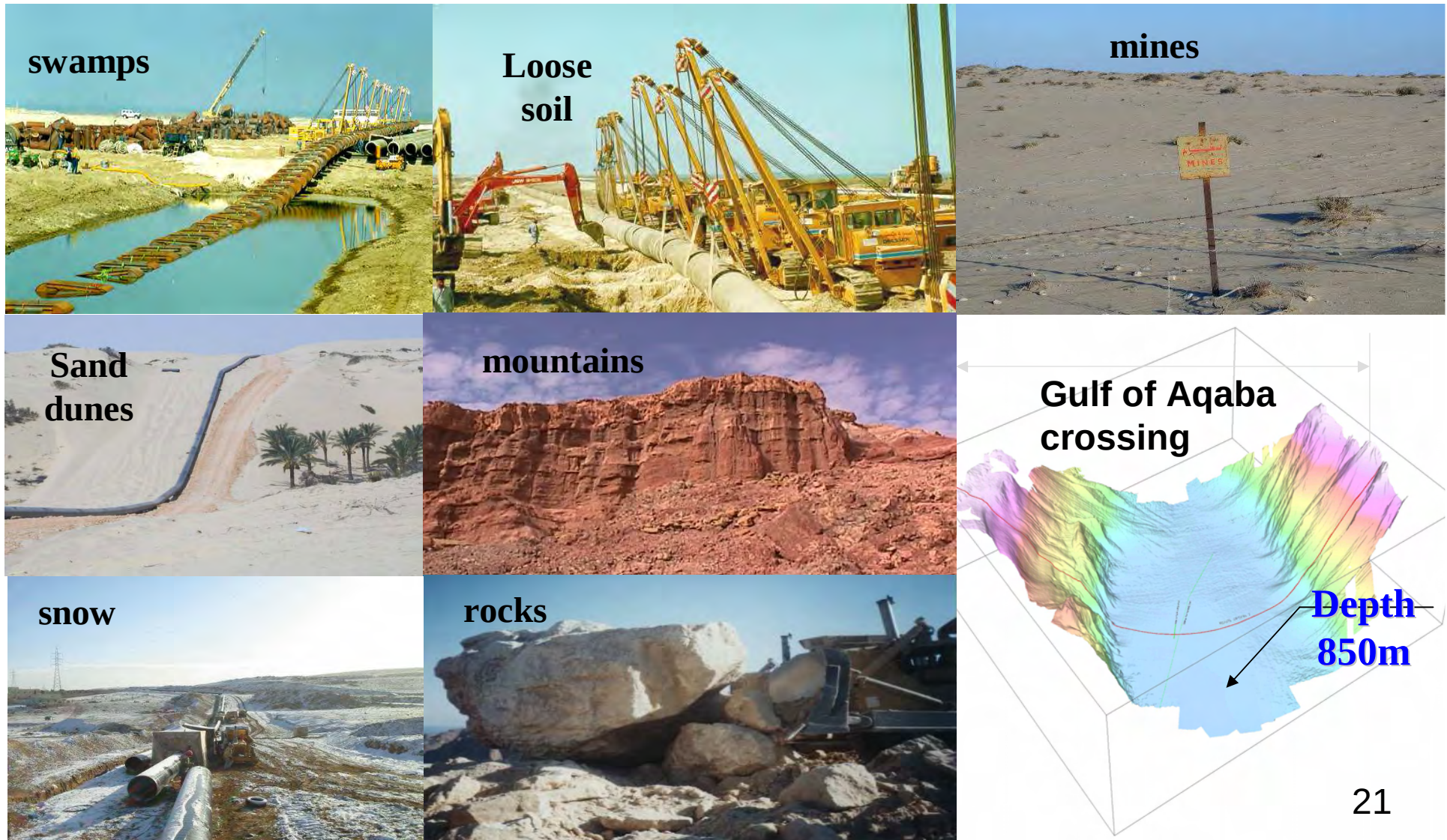


Key success factors

1. Political will and support



2- Completion of phases 1 &2 in record time due to efforts of dedicated consortium of Egyptian petroleum sector companies



3- Securing of required Finance:

The projects are economically feasible, a number of reputable financial institutions have responded swiftly to finance a portion of the completed phases. For phases 1 and 2, **430 million US dollars**, were provided by the following financial institutions :

- **European Investment Bank**
(120 MMUS\$)
- **The Kuwait Fund For Arab Economic Development**
(100 MMUS\$)
- **The Arab Fund For Economic And Social Development**
(50 MMUS\$)
- **The Jordanian Housing Bank for Trade and Finance
(Commercial Bank)**
(160 MMUS\$)

EU Support



PROJECTS OF
PAN-EUROPEAN
INTEREST

PROPOSED PRIORITY AXES
FOR NATURAL GAS PIPELINES

The EU's Vision

The 3rd Euro-Mediterranean Energy Ministerial Meeting- Athens (2002)



The Euro-Med Energy Ministerial Conference in Rome in December 2003

During the Euro-Med Energy Ministerial Conference in Rome in December 2003, two Declarations of Intent were signed to study the extension of the Arab Gas Pipeline and define a gas pipeline corridor to Central Europe.

Egypt as An Energy Hub

“A New Horizon”

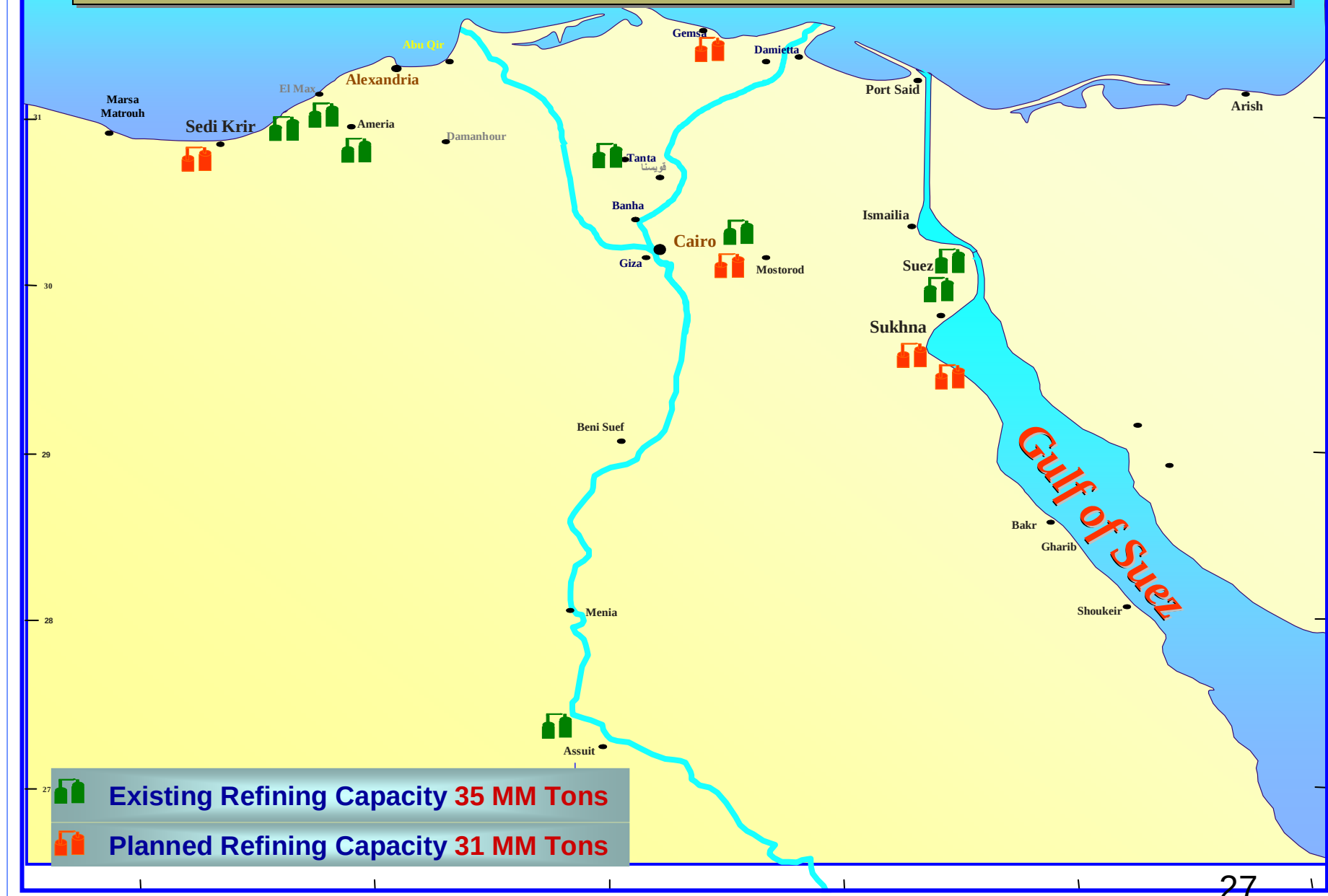
Egypt as An Energy Hub

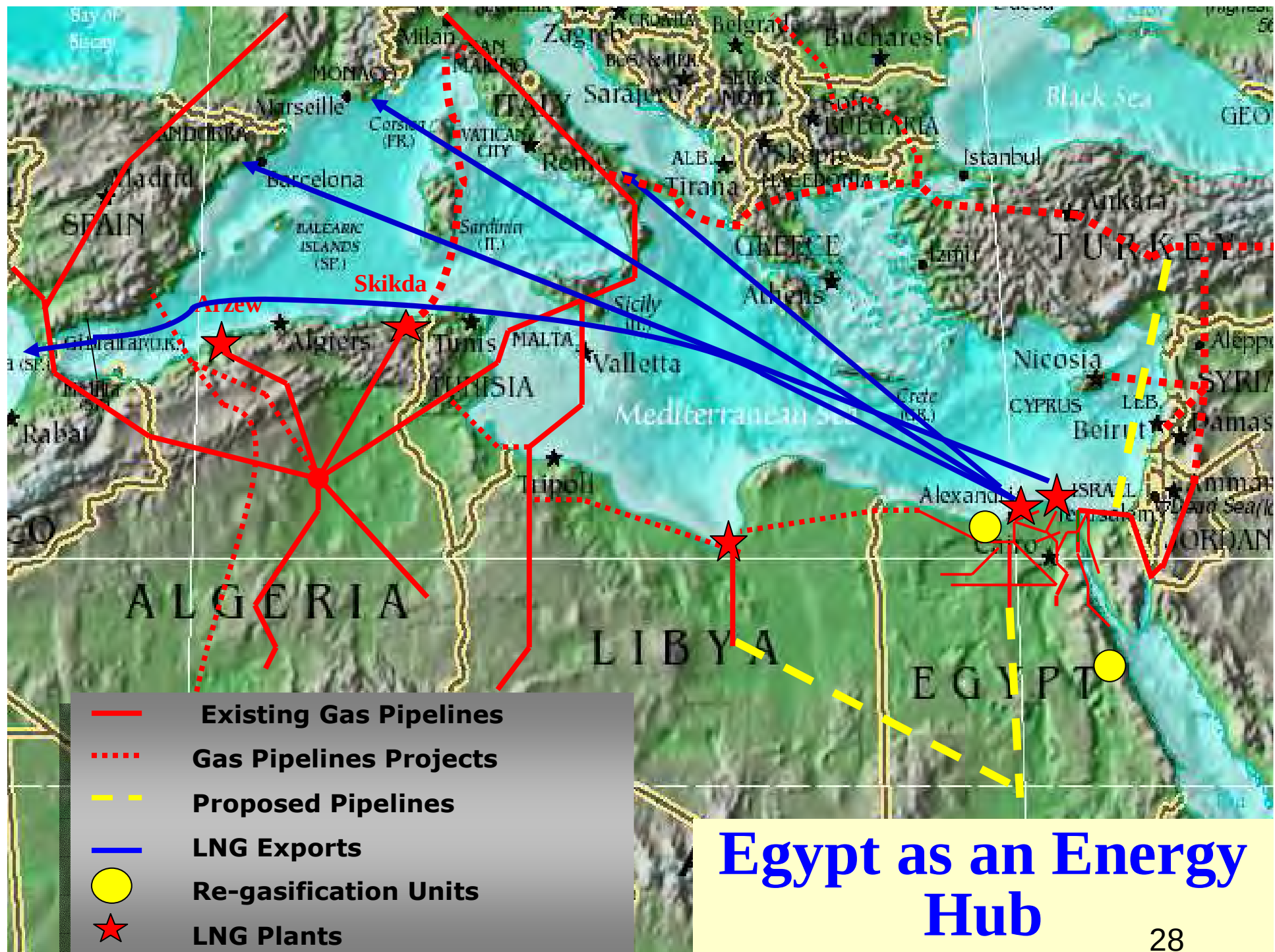
- **Egypt, with its strategic location can be a hub for oil and gas playing a major role in securing part of the Euro-Med energy supplies.**
- **This vision will help to achieve the EU's objective to improve Energy Security and Supply Diversification.**

Egyptian Assets

- **Political And Economical Stability.**
- **Appropriate Fiscal Incentives.**
- **Availability Of Skilled Human Capital.**
- **Transparent Regulations.**
- **Positive Relationship Between Petroleum Sector And Our Partners.**
- **The Success Already Achieved This Far Coping With Our Planned Strategic Goals.**

New Planned Refineries





Thank You

